

AHAMMAD SHAWKI

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EDUCATION

Bangladesh University of Engineering and Technology (BUET)

BSc in Computer Science and Engineering

Expected 2028

Dhaka, Bangladesh

Coursework: Data Structures and Algorithms I and II, Database, Software Engineering, Computer Architecture, Theory of Computation.

EXPERIENCE

Founder and Developer, SrotDev

May 2025 to Present

- Lead an engineering collective shipping production web and mobile applications across React, Next.js, Django, FastAPI and PostgreSQL.
- Architected CareForAll as seven independently deployable Django services behind an API gateway, with its own Prometheus and Grafana observability stack.
- Built Srabon, a gamified science learning platform with multilingual and accessibility support, as the team's flagship product.

President of Web and App Development, NDITC

Mar 2022 to Jun 2024

- Led a 30-person engineering team; delivered the rebuilt official website and the UI/UX for the club's native mobile application.
- Raised over 1,000,000 BDT in sponsorship across 100+ company negotiations for INIT, the club's national annual event.
- Taught web development fundamentals directly to more than 500 students as a workshop instructor.

Founder and Team Lead, The AS8 Organization

Feb 2021 to May 2022

- Built a developer community that reached more than 20,000 learners through free programmes, courses and technical content.
- Partnered with Codecademy, Hack Club and DeepLearning.AI; wrote articles published on freeCodeCamp News.

PROJECTS

HALFSPREAD, autonomous options trading agent

Python, Alpaca, GitHub Actions

- Prices execution cost before every order, having measured bid-ask cost from 0.52% to 33% of contract value depending on time to expiry. Every published figure re-derives from an append-only journal with no API key and no network, and the build fails if one stops reproducing.

Cassandra, autonomous spreadsheet repair

Google ADK, Gemini, Cloud Run, Firestore

- Agent fleet that repairs spreadsheet defects and proves each fix by recalculating the workbook, rejecting any correction that moves the wrong cell. Repaired five defects in 114 seconds unattended on a model reporting \$6.2M in income for a company that had lost \$1.7M.

DeepSIFT, auditable digital forensics

MCP, Python, Volatility 3, MITRE ATT&CK

- 148-tool Model Context Protocol layer that parses forensic binaries into structured JSON before the model reads them, with a signable chain of custody. Scored 4/4 against published ground truth on two cases, zero hallucinations, 100% claim grounding.

MACHINE LEARNING COMPETITIONS

CADENCE, voice-based Parkinson's screening

ML Empowerment Build Challenge, 2026

- Won best overall project. Screens from a 30-second voice sample and explains every score, validated across three independent language corpora using cross-database testing and domain-adversarial training to avoid microphone bias.

Olikobocho, Bengali LLM hallucination detection

IUTCS Datathon 2.0, 2026

- Third on the public leaderboard at 0.922 F1 on the hallucinated class. A decision ladder resolves most rows deterministically against source corpora and an 89k-question bank, sending only the residual to an open-weight judge with Wikipedia retrieval.

DL Sprint 4.0, Bangla long-form speech recognition

BUET CSE Fest, 2026

- Transcription of hours-long Bangla audio with overlapping speakers, background music and long silences, scored on word error rate weighted by sentence length.

AWARDS

Grand Prize Winner, FLIR App Challenge 2025 to 2026, for SolarSnap. **Best Overall Project**, ML Empowerment Build Challenge 2.0, 2026, for CADENCE. **Best ERNIE Multimodal Application**, ERNIE AI Developer Challenge by Baidu, 2026, for Doclyst. **Honourable Mention**, LMA EDGE Hackathon, 2026, presented at the London finale, for Coven. Four wins across 32 hackathons entered.

TECHNICAL SKILLS

Languages: Python, C/C++, TypeScript, JavaScript, Java, SQL, Dart, HTML/CSS

Frameworks: React, Next.js, Django, Django REST Framework, FastAPI, Flask, Node.js, Flutter, JavaFX

AI and Machine Learning: PyTorch, scikit-learn, Model Context Protocol, Google ADK, Retrieval Augmented Generation, Whisper, Gemini, Claude, Pandas, NumPy

Infrastructure: Docker, Google Cloud (Cloud Run, Pub/Sub, Firestore, Vertex AI), AWS (Bedrock, ECS, Lambda), PostgreSQL, MongoDB, Redis, GitHub Actions, Grafana